

Kaggle Data Science Survey Trends

A five-year view of participation, geography, roles, gender representation, and programming-language usage (2017-2021).

106,301

survey responses

5

survey years

55%

response growth

84%

used Python in 2021

Source

A merged public dataset built from Kaggle's annual Machine Learning and Data Science surveys for 2017 through 2021.

Interpretation

Findings describe voluntary Kaggle survey respondents. They are not population estimates for the global workforce.

Executive summary

Survey participation more than doubled across the available period. India and the United States supplied the largest 2021 response counts. Python remained the dominant regularly used programming language, while women remained underrepresented at 18.8% of 2021 respondents. Students were the largest reported role group.

Data and method

Annual coverage

Year	Responses	Countries	Median completion time
2017	16,716	52	Not available
2018	23,859	58	17.0 min
2019	19,717	59	9.0 min
2020	20,036	55	10.4 min
2021	25,973	66	10.9 min

Cleaning decisions

- Removed the embedded question-description row.
- Retained identical answer profiles because no respondent ID is available.
- Harmonized Male/Man and Female/Woman labels.
- Repaired known spreadsheet date conversions in team-size ranges.
- Corrected visible UTF-8 mojibake in analytical fields.
- Preserved missing answers instead of treating them as zero.

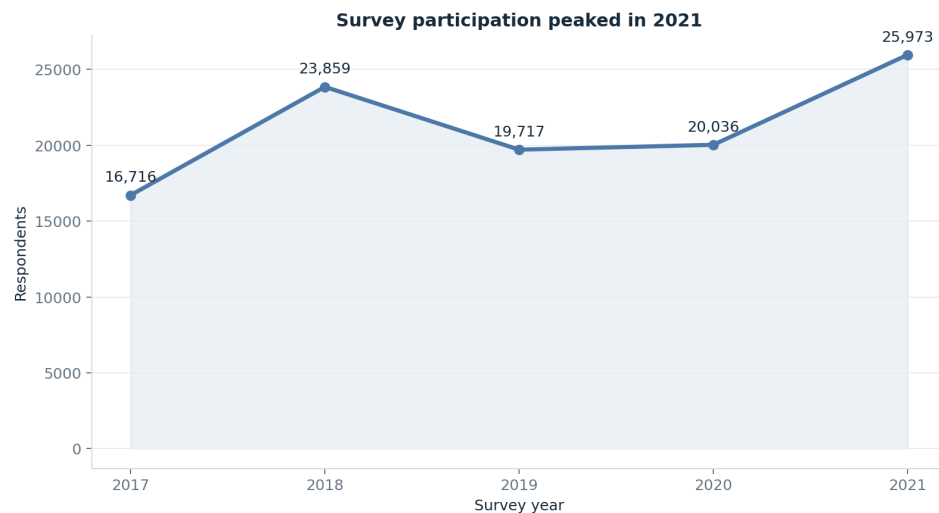
Comparability rules

Questionnaire content and labels changed between years. Trend charts use fields with stable meaning, and multi-select language shares use all respondents in each year as the denominator. Missing values can reflect non-response, skip logic, or a question not being asked; they are not interpreted as negative answers.

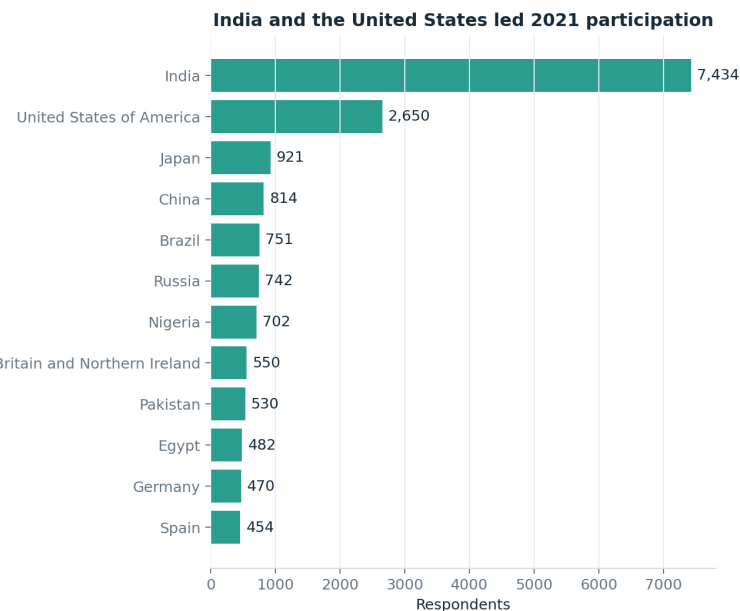
Analytical scope

This report emphasizes descriptive patterns that can be reproduced from the supplied merged file. It does not estimate causal effects, project the workforce, or compare compensation across changing salary bands.

Participation and geography

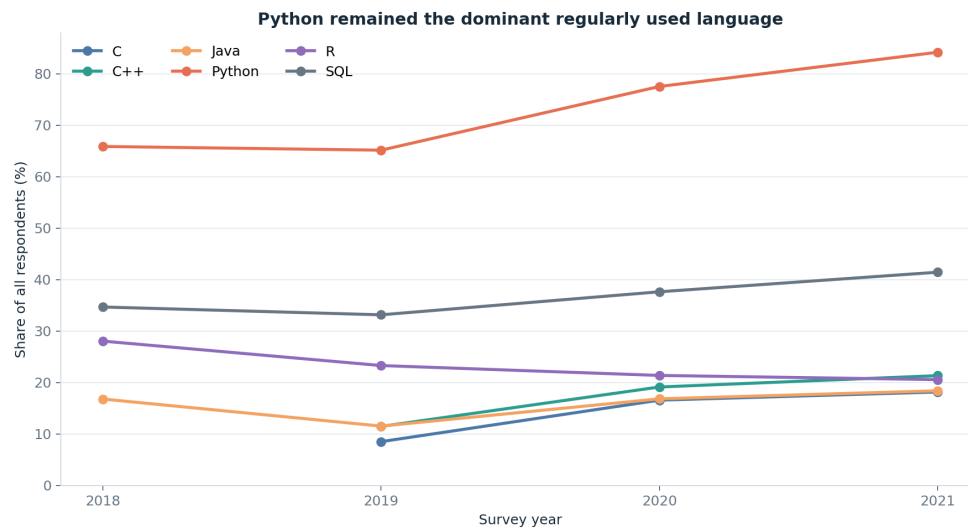


Responses increased from 16,716 in 2017 to 25,973 in 2021. Growth in participation should not be read as workforce growth.

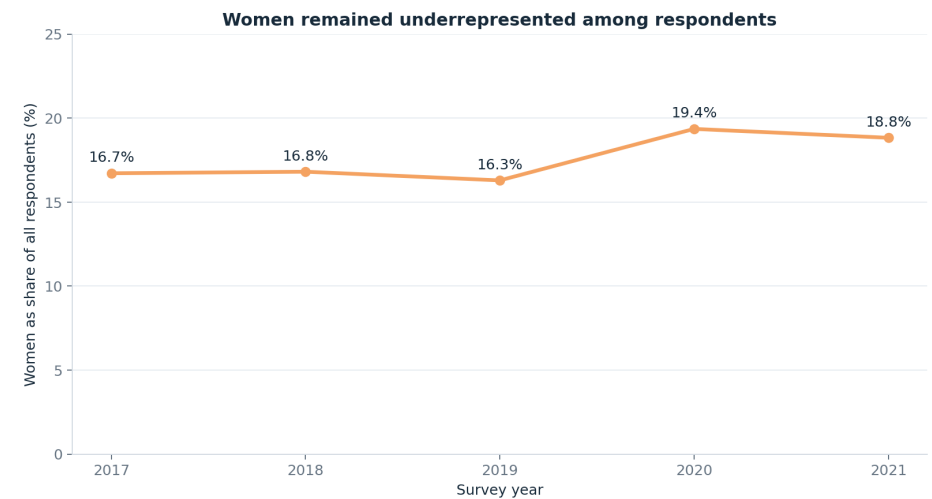


India contributed 7,434 responses in 2021, followed by the United States with 2,650. Country rankings describe the survey sample, not workforce size.

Languages and representation

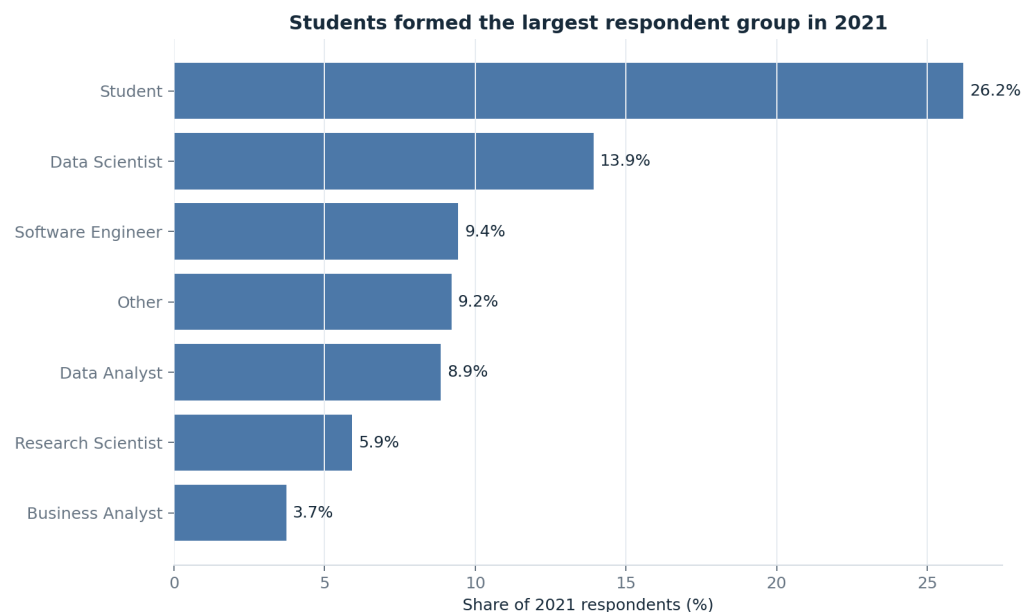


Python was the dominant regular-use language in every year with available multi-select fields (2018-2021). The 2019 dip appears across several languages, reinforcing the need to interpret annual shares alongside questionnaire and respondent-mix changes.



Women's share was 16.6% in 2017 and 18.8% in 2021, with a peak in 2020. Representation improved, but the gap remained substantial.

Roles and final takeaways



What the data supports

- Participation expanded materially across the five available surveys.
- Geographic participation was concentrated, especially in India.
- Python held a clear lead in regular language use.
- Students were the largest 2021 role group at 26.2%.
- Gender representation improved modestly but remained imbalanced.

What the data does not support

- Claims about the entire data-science workforce.
- Causal explanations for year-to-year changes.
- Population-level estimates for the global data-science workforce.
- Direct compensation trends without additional band harmonization.

Reproducibility

The repository includes the compressed source, processed extracts, a runnable Jupyter notebook, chart files, and detailed methodology notes.